EXECUTIVE SUMMARY

EduPro Instructor Performance and Course Quality Evaluation

1. Overview

The EduPro Instructor Performance and Course Quality Evaluation project applies data analytics to assess instructor performance, course quality, teaching experience, and platform transaction activity. Its purpose is to provide a transparent, evidence-informed framework for educational quality assurance and administrative decision-making.

The project integrates teacher, course, and transaction datasets using Python, statistical analysis, data visualization, and a Streamlit dashboard. It is designed to help government stakeholders and education administrators identify patterns, investigate quality concerns, and prioritize improvement initiatives.

The reported analysis covers 60 teachers, 60 courses, and 10,000 transactions. The findings provide an initial baseline for further investigation rather than a definitive assessment of educational effectiveness.

2. Key Findings

Performance indicator	Reported result
Number of teachers	60
Number of courses	60
Number of transactions	10,000
Average teacher rating	3.125/5
Average course rating	3.098/5
Average teaching experience	6.28 years
Experience vs. teacher rating	Positive association
Experience vs. course rating	Weak, statistically insignificant association

Finding 1: Teaching experience is positively associated with teacher ratings.

Pearson's correlation coefficient is 0.598, while Spearman's rank correlation coefficient is 0.671. Both reported relationships are statistically significant at the conventional 5% level. This suggests that more experienced teachers tend to receive higher teacher ratings in the observed dataset.

However, this result does not prove that experience causes higher ratings. Other factors, including subject expertise, communication, course assignments, and learner expectations, may influence the relationship.

Finding 2: Teaching experience does not significantly predict course ratings in the reported analysis.

The Pearson correlation between experience and associated course ratings is −0.138, with an approximate p-value of 0.293. The relationship is weak and not statistically significant at the 5% level.

This finding supports evaluating course quality independently from instructor experience. Course design, learning materials, assessment methods, accessibility, and learner support should be examined directly.

Finding 3: Average ratings provide a baseline, not a complete quality assessment.

The average teacher rating is 3.125/5, and the average course rating is 3.098/5. These values provide a starting point for monitoring performance. Further analysis of rating distributions, response counts, course categories, course levels, and learner outcomes is required to determine where improvement is needed.

Finding 4: Transaction activity is useful for operational monitoring but is not a direct measure of learning effectiveness.

The dataset contains 10,000 transactions. These records can support analysis of platform activity and potential demand. However, transaction volume should only be treated as an enrollment proxy after confirming what each transaction

represents. It should not be interpreted as proof of learning achievement or teaching quality.

3. Strategic Implications for Government

The findings have four primary implications for educational governance.

First, performance evaluation should be multidimensional. Instructor ratings, course ratings, teaching experience, learner satisfaction, and learning outcomes measure different aspects of the educational process. A single score should not determine an instructor's overall performance.

Second, professional development should be evidence-informed. The positive relationship between experience and teacher ratings justifies further investigation of teaching practices, mentoring, and peer learning. It does not justify assuming that less-experienced teachers are necessarily less effective.

Third, course quality requires independent oversight. Educational authorities should evaluate curriculum relevance, content accuracy, instructional design, assessment alignment, accessibility, and learner support alongside instructor performance.

Fourth, data-driven governance requires safeguards. Transparent KPI definitions, validated datasets, adequate sample sizes, and human review are essential before analytical findings are used for high-impact decisions.

4. Recommended Policy Actions

1. **Establish a standardized educational quality framework.** Define the indicators, calculation methods, reporting frequency, and interpretation rules used to assess instructors and courses.
2. **Introduce structured instructor development.** Use performance indicators to identify potential mentoring needs and professional-development opportunities.
3. **Conduct periodic course-quality audits.** Review courses with persistently low ratings, adequate response counts, or other validated quality concerns.
4. **Strengthen educational data infrastructure.** Improve the collection of enrollment records, completion rates, assessment results, rating counts, learner feedback, and instructor-course assignment information.
5. **Adopt transparent dashboard reporting.** Provide authorized administrators with summaries of performance indicators, category-level comparisons, data-quality checks, and trends over time.
6. **Protect fairness in performance decisions.** Avoid automatic rankings or disciplinary action based solely on unadjusted ratings or transaction counts. Require contextual review and additional evidence.
7. **Measure improvement outcomes.** Assess whether mentoring, course redesign, and other interventions lead to improvements in learner satisfaction, completion, assessment performance, and educational outcomes.

5. Proposed 90-Day Implementation Plan

Phase 1: Validation — Days 1–30

- Verify all reported statistics against the original CSV datasets.
- Confirm the meaning of transaction records and whether they represent unique enrollments.
- Validate teacher-course relationships and rating counts.
- Standardize KPI definitions and document data limitations.
- Recalculate correlations and confidence intervals from the raw data.

Expected output: A validated baseline report with documented definitions and data-quality findings.

Phase 2: Diagnostic Review — Days 31–60

- Examine rating distributions across expertise areas, course categories, and course levels.
- Identify courses requiring further investigation using appropriate sample-size thresholds.
- Review learner feedback and relevant course materials.
- Investigate factors that may influence rating differences.

Expected output: A prioritized list of improvement areas supported by quantitative and qualitative evidence.

Phase 3: Improvement Pilot — Days 61–90

- Pilot targeted instructor mentoring and course-improvement plans.
- Establish baseline measures for each intervention.
- Monitor changes in ratings and, where available, completion and assessment outcomes.
- Prepare a follow-up report describing progress, limitations, and recommended next steps.

Expected output: An initial evaluation of improvement activities and a recommendation on whether to expand the pilot.

6. Risks and Limitations

The reported results should be interpreted cautiously.

- The teacher-level analysis involves 60 teacher records, not 10,000 independent observations.
- Correlation does not establish causation.
- Ratings may be influenced by response bias, course difficulty, unequal response counts, and learner characteristics.
- Course ratings cannot necessarily be attributed to an individual teacher when multiple teachers are associated with a course.
- Transaction volume may not represent unique enrollments or learning achievement.

- The summary statistics have not been independently recalculated from the raw datasets for this document.

These limitations do not eliminate the value of the project. They establish the conditions under which the results can be used responsibly.

7. Expected Benefits

If implemented with appropriate validation and oversight, the EduPro evaluation framework can support:

- More transparent monitoring of educational quality.
- Earlier identification of potential instructor and course improvement needs.
- Better targeting of professional-development resources.
- Consistent reporting of educational performance indicators.
- Improved integration of quantitative analysis into administrative planning.
- More systematic evaluation of quality-improvement initiatives.

These are expected benefits of the proposed framework, not outcomes that have already been demonstrated by the reported analysis.

8. Decision Requested

Government and education-platform stakeholders are requested to approve a time-limited validation and quality-improvement pilot.

The pilot should prioritize data verification, standardized performance indicators, independent course-quality assessment, instructor development, and measurement of learner outcomes. Any subsequent expansion should be based on a validated follow-up evaluation.

9. Conclusion

The EduPro project establishes an initial analytical foundation for monitoring instructor performance and course quality. Its reported results show a positive association between teaching experience and teacher ratings, while the relationship between experience and course ratings is weak and statistically insignificant.

The principal policy recommendation is to adopt a multidimensional, transparent, and evidence-informed quality-assurance framework. Instructor ratings should inform—but not determine—performance decisions, and course quality should be assessed through direct evaluation of educational design and learner outcomes.

With validated data, clear governance rules, and continuous monitoring, the proposed framework can support more consistent educational oversight and targeted quality improvement.

Prepared for: Government stakeholders, education administrators, and digital learning platform managers.

Project: EduPro Instructor Performance and Course Quality Evaluation.

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